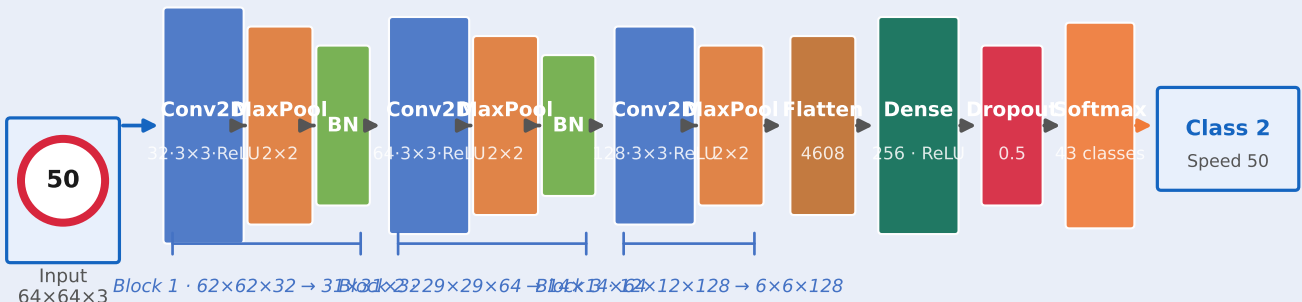


Network Architectures — Traffic Sign Recognition Framework

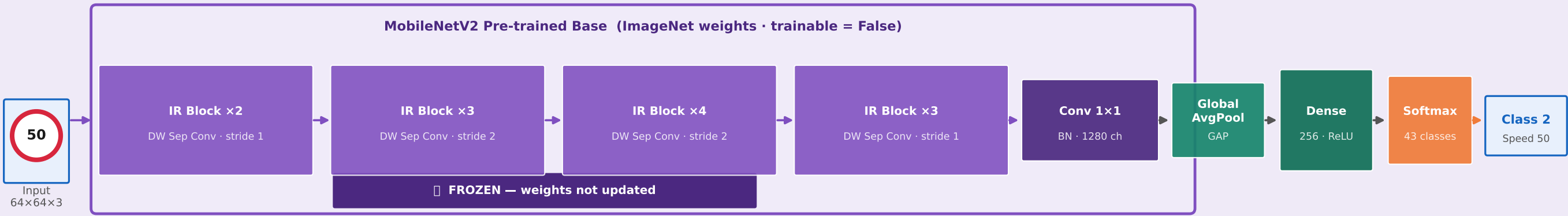
GTSRB · 43 Classes · Input 64×64×3 RGB

Custom CNN



Loss: Categorical Cross-Entropy · Optimiser: Adam (lr=0.001) · Max Epochs: 25 · EarlyStopping patience=5 · restore_best_weights=True

MobileNetV2
Transfer Learning



Loss: Categorical Cross-Entropy · Optimiser: Adam (default lr=0.001) · Epochs: 10 · Base trainable=False · ImageNet pre-trained weights

- Conv2D + ReLU

MaxPooling2D

Batch Normalisation

Dense (FC)
- Dropout

Softmax Output

Inverted Residual (DW Sep Conv)

Global Avg Pooling